

CORE COURSE XIII: 6B13CSC COMPILER DESIGN

SEMESTER	COURSE CODE	HOURS PER WEEK	CREDIT	EXAM HRS
6	6B13CSC	4	4	3

COURSE OUTCOME

- CO1:** Learn the basic principles of compiler.
CO2: Get an idea about the related programs.
CO3: Understand different components of a compiler.
CO4: Understand the phases of a compiler.

Unit I: INTRODUCTION TO COMPILING

Compilers, Analysis of the Source program, phases of a compiler, cousins of the compiler, grouping of phases, compiler construction tools.

(18Hrs)**Unit II: LEXICAL ANALYSIS**

Role of Lexical Analyzer, Input buffering, Specification of tokens, recognition of tokens, Finite Automata.

(18Hrs)**Unit III: SYNTAX ANALYSIS**

The role of a Parser, context free grammars, Top down parsing, Recursive Descent Parsing, Predictive Parsers, bottom up parsing, shift reduce parsing, operator precedence parsing.

(18Hrs)**Unit IV: CODE GENERATION & OPTIMIZATION**

Symbol table, Intermediate languages, Issues in the design of code generator, the target machine, basic blocks and flow graphs, peep-hole optimization, principal sources of optimization, optimization of basic blocks, Loops in flow graphs

(18Hrs)**Books for Study:**

1. Alfred V Aho, Ravi Sethi & Jeffrey D Ullman, "Compilers- Principles, Techniques and Tools", Pearson education

Books for Reference:

1. KVN Sunitha, Compiler Construction, Pearson Education

2. Parag H Dave, Himanshu B Dave, Compilers –Principles and Practice

Marks including choice:

Unit	Marks
I	15
II	15
III	15
IV	15